(WEAK) SOLUTIONS DESCEND ALONG THE INTERMEDIATE EXTENSION OF DISCRETE VALUATION RINGS AND COMPOSE

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Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file more-algebra.tex; no statement or proof step was rewritten, reordered or removed. The counted statement is the labeled lemma environment at source lines 34516-34527 (canonical tag 09ES) and the counted proof is the proof environment at source lines 34529-34552, which immediately follows it and is its own proof.

Editorial note. The proof ends with the sentence that a similar argument works for the second assertion; that sentence is the author's own and is preserved verbatim inside the counted span, the second case being the same localization argument applied to the compositional hypothesis.

Editorial note. The two references are to earlier results of this chapter, the construction remark and the formally smooth criterion; both are navigation stubs with their roles recorded in the reasons.

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Allowed standard prerequisites (editorial).

- The reduced base change of a finite compositum is a finite product of finite extensions and the corresponding map of integral closures is a product of the maps considered in the construction remark, hence surjective on spectra. invoked at proof lines 34529-34538. Provides a maximal ideal of the larger integral closure lying over a chosen prime of the smaller one, which is the point at which the comparison is made.
- Weak unramifiedness for a composite of extensions of discrete valuation rings implies weak unramifiedness of the intermediate extension, and the same holds for separability of the residue field extensions. used at proof lines 34538-34547. Yields assertion (1) by comparing the composite with its first factor.
- The equivalence between weak unramifiedness with separable residue extension and formal smoothness in the adic topology, which is the definition of a weak solution. invoked at proof lines 34547-34549. Translates the conclusion about weak unramifiedness and separable residue fields back into the statement about weak solutions.

Date: stacks-project:more-algebra.tex:local-snapshot-20261012; GNU FDL 1.2 or later.

Source reference. Original remark 09EM: sources/stacks-more-algebra/pinned-original-source.tex, source lines 33882–33882. The reference identifies the map between the integral closures as a finite product of the standard maps, used at proof lines 34533-34538 to get surjectivity on spectra; the localization argument built on it is inside the counted span.

Source reference. Original lemma 09E7: sources/stacks-more-algebra/pinned-original-source.tex, source lines 32989–32989. The reference is invoked at proof lines 34547-34549 to translate the conclusions about weak unramifiedness and separable residue extensions into the statement about (weak) solutions; it is the last step and everything leading to it is inside the counted span.

Lemma 0.2. *Let $A \to B \to C$ be extensions of discrete valuation rings with fraction fields $K \subset L \subset M$. Let K_1/K be a finite extension.*

1. *If K_1 is a (weak) solution for $A \to C$, then K_1 is a (weak) solution for $A \to B$.*
2. *If K_1 is a (weak) solution for $A \to B$ and $L_1 = (L \otimes_K K_1)_{red}$ is a product of fields which are (weak) solutions for $B \to C$, then K_1 is a (weak) solution for $A \to C$.*

Proof. Let $L_1 = (L \otimes_K K_1)_{red}$ and $M_1 = (M \otimes_K K_1)_{red}$ and let $B_1 \subset L_1$ and $C_1 \subset M_1$ be the integral closure of B and C. Note that $M_1 = (M \otimes_L L_1)_{red}$ and that L_1 is a (nonempty) finite product of finite extensions of L. Hence the ring map $B_1 \to C_1$ is a finite product of ring maps of the form discussed in Remark 09EM. In particular, the map $\mathrm{Spec}(C_1) \to \mathrm{Spec}(B_1)$ is surjective. Choose a maximal ideal $\mathfrak{m} \subset C_1$ and consider the extensions of discrete valuation rings

$$(A_1)_{A_1 \cap \mathfrak{m}} \to (B_1)_{B_1 \cap \mathfrak{m}} \to (C_1)_{\mathfrak{m}}$$

If the composition is weakly unramified, so is the map $(A_1)_{A_1 \cap \mathfrak{m}} \to (B_1)_{B_1 \cap \mathfrak{m}}$. If the residue field extension $\kappa_{A_1 \cap \mathfrak{m}} \to \kappa_{\mathfrak{m}}$ is separable, so is the subextension $\kappa_{A_1 \cap \mathfrak{m}} \to \kappa_{B_1 \cap \mathfrak{m}}$. Taking into account Lemma 09E7 this proves (1). A similar argument works for (2). $\square$